

The Price-Tag Decision: Logistic Regression and Menu Costs at SunCoast Retail Mart

ISM 6642 Machine Learning (MSIS), Assignment 6 | Giuseppe Laricchia | Florida International University | September 29, 2026

Code: `retail_relabel_analysis.py / .ipynb` (runs end-to-end from `retail_relabel.csv`; split seed `random_state = 7`). Every number below is written to `outputs/results.json` by that code.

Part 1. Feature engineering and exploratory analysis

The data hold 1,500 SKUs with no missing values or duplicate SKUs. I built **CompAvg** (mean of the BigBox Depot, ValueMart and QuickShop prices), the signed gap **GapPct** = $100 \cdot (\text{StorePrice} - \text{CompAvg}) / \text{CompAvg}$, its magnitude **AbsGapPct** = $|\text{GapPct}|$, **LogUnits** = $\ln(\text{WeeklyUnitsSold})$ and **MarginPct** = $100 \cdot (\text{StorePrice} - \text{UnitCost}) / \text{StorePrice}$. The store averages +3.6% versus competitors (SD 12.7 pts); 59% of SKUs are priced above the competitor average and the rest below. The typical misalignment is large: mean $|\text{gap}|$ 11.0%, median 10.1%.

The overall relabel rate is **40.0%**. By category the rate is narrow, from 37.8% (Household) to 44.0% (Electronics), so category alone does not drive the decision. Tag technology does: **ESL shelves relabel 54.1%** of items against **33.9% for paper tags**, and an ESL change costs \$0.06 on average against \$1.45 for paper (about 24x).

Menu-cost interpretation. The two groups are equally mispriced (mean $|\text{gap}|$ 10.8% ESL vs. 11.1% paper), so the difference is not that ESL items need more correcting. It is that the correction is cheaper. With a paper tag, a small misprice is not worth the labor, so the store tolerates a wider band of inaction. Figure 1 and the gap-band table in the code confirm this: in every $|\text{gap}|$ band ESL items relabel more often, and the difference is largest for small gaps (0–5%: 23% ESL vs. 9% paper; 20%+: 88% vs. 74%). Once the gap is large enough, both tag types get fixed; the menu cost mostly decides the fate of the moderate misprices.

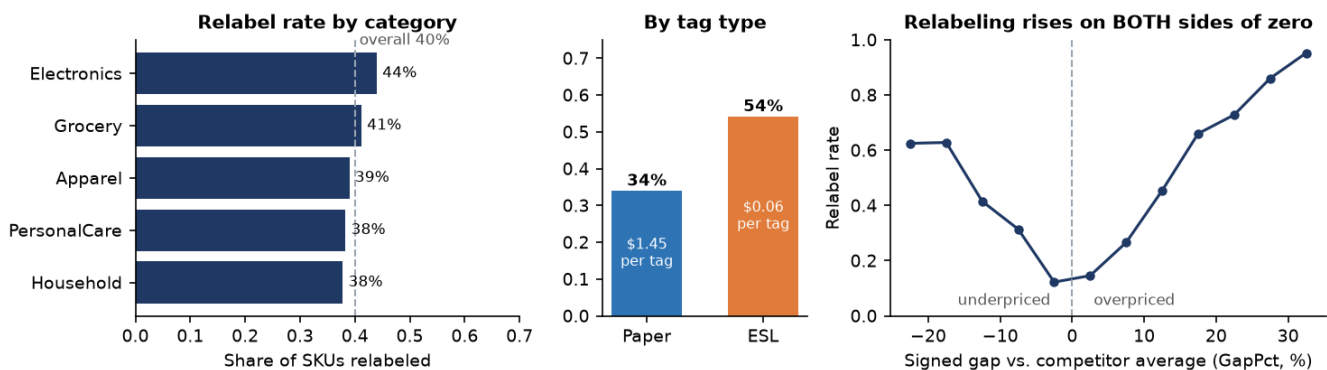


Figure 1. Relabel rate by category (left), by tag type (centre), and by signed price gap in 5-point bins (right).

Part 2. A naive model with the signed gap

Model 1 ($\text{Relabel} \sim \text{GapPct} + \text{UnitCost} + \text{RelabelCost}$) has **pseudo $R^2 = 0.077$** and **AIC = 1,871.0** (BIC 1,892.3). GapPct is positive and significant ($b = 0.038$, $p < 0.001$), RelabelCost is negative ($b = -0.639$, $p < 0.001$) and UnitCost is not significant ($p = 0.88$). The model "works" statistically but explains less than 8% of the log-likelihood.

Why the signed gap is the wrong functional form. An item 10% under competitors and an item 10% over are both mispriced. The first gives away margin on every unit; the second loses volume to competitors. The owner's response is the same in both cases: change the tag. So the probability of relabeling should be lowest near a zero gap and rise in **both** directions (a V shape). A logit in GapPct can only fit an S-curve that rises in one direction, so it treats a deeply underpriced item as the one least in need of a new tag. The data show the V clearly (Figure 1, right): 50% of items $\geq 10\%$ under competitors were relabeled ($n = 257$), only 13% of items within $\pm 5\%$, and 63% of items $\geq 10\%$ over ($n = 503$). Model 1 instead predicts 28% at -10% , 36% at 0% and 45% at $+10\%$: the ranking is wrong on the underpriced side. The positive slope survives only because 59% of SKUs sit on the overpriced side of the V; the underpriced side pulls against it. That is why Relabel correlates only 0.21 with GapPct but 0.46 with $|\text{GapPct}|$. Changing only this one variable (Model 1 with AbsGapPct) raises pseudo R^2 from 0.077 to 0.221 and cuts AIC from 1,871 to 1,582. A signed variable that should have been measured symmetrically diluted a real effect.

Part 3. The economically motivated model (Model 2)

Predictor	Coef.	Odds ratio	95% CI (OR)	z	p
AbsGapPct (per pt)	0.1778	1.195	[1.169, 1.220]	16.25	< 0.001
CostChangePct (per pt)	0.1276	1.136	[1.099, 1.174]	7.58	< 0.001
LogUnits (per 1 log unit)	0.4324	1.541	[1.192, 1.992]	3.30	< 0.001

RelabelCost (per \$1)	-0.8589	0.424	[0.358, 0.501]	-10.08	< 0.001
DaysSinceLastChange (per day)	0.0040	1.004	[1.003, 1.005]	6.37	< 0.001
UnitCost (per \$1)	0.0004	1.000	[0.993, 1.008]	0.10	0.919

Plain-English reading for the owner (holding the other factors constant):

- **AbsGapPct:** each extra percentage point of misalignment with competitors multiplies the odds of changing the tag by 1.19 (about +19%); a 5-point larger gap multiplies them by 2.43. The bigger the misprice, the more likely you fix it. This is the strongest effect in the model ($z = 16.2$).
- **RelabelCost:** each extra \$1 it costs to change a tag cuts the odds of changing it by 58% (OR 0.42). The average paper tag costs about \$1.39 more than an ESL change, which lowers the odds by about 70%. Expensive tags leave misprices on the shelf.
- **LogUnits:** doubling an item's weekly unit sales multiplies the odds by 1.35 (+35%). Fast movers get repriced more often because the fixed tag cost is earned back on more units.

Why CostChangePct matters and UnitCost does not. CostChangePct has OR 1.136 per point ($p < 0.001$; a 5-point supplier increase multiplies the odds by 1.89). UnitCost has OR 1.0004 (95% CI 0.993–1.008, $p = 0.92$). This is what the economics predicts. A tag is changed when the current price has become wrong, so only something that has moved since the last pricing can trigger a change. The level of unit cost was already built into the current price when it was set: a \$60 item was priced knowing it costs \$60, so being expensive gives no reason to reprice today. A supplier cost change moves the optimal price away from the posted one and squeezes margin; that is a new wedge that must be closed. In the data, relabel rates by UnitCost quartile are flat (39%, 43%, 38%, 41%), and UnitCost is almost uncorrelated with CostChangePct ($r = -0.03$), so the null is not an artifact of collinearity. The same logic explains DaysSinceLastChange (OR 1.44 per 90 days): drift accumulates on stale tags.

Model	Params	Pseudo R ²	AIC	BIC
Model 1: GapPct + UnitCost + RelabelCost	4	0.077	1,871.0	1,892.3
Model 1 with AbsGapPct instead of GapPct	4	0.221	1,581.6	1,602.9
Model 2 (six economic predictors)	7	0.275	1,477.8	1,515.0
Model 2 + GapPct (direction check)	8	0.275	1,479.8	1,522.3
Model 2 + MarginPct	8	0.275	1,479.8	1,522.3

Model choice. Model 2 raises pseudo R² from 0.077 to 0.275 and lowers AIC by 393 and BIC by 377 points. Differences above 10 are already decisive, and BIC's heavier penalty for Model 2's three extra parameters does not change the ranking. **I would deploy Model 2.** Two checks support it: adding the signed GapPct back to Model 2 adds nothing ($p = 0.87$, AIC rises), so once misalignment is measured symmetrically its direction carries no extra information. Adding MarginPct also adds nothing ($p = 0.89$); it is 0.90 correlated with GapPct, so it is mostly the signed gap again, and a margin level, like a cost level, is not a trigger for repricing.

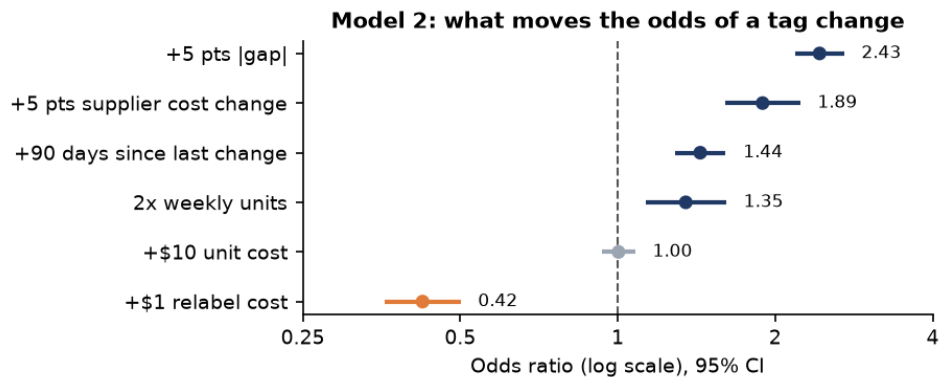


Figure 2. Model 2 odds ratios scaled to meaningful changes, with 95% CIs (grey = not significant).

Part 4. Predictive evaluation (out of sample)

A stratified 70/30 split (random_state = 7) gives 1,050 training and 450 test SKUs, each with a 40.0% relabel rate. Refit on the training data, Model 2's odds ratios barely move (e.g. AbsGapPct 1.183 vs. 1.195 on the full data; UnitCost stays insignificant). At a 0.5 cutoff the test results are:

	TN	FP	FN	TP	Accuracy	Precision	Recall	Specificity	AUC
Test set (n = 450)	233	37	64	116	77.6%	75.8%	64.4%	86.3%	0.853
Training set (n = 1,050)					76.5%				0.821

Test **accuracy is 77.6%** against a 60.0% baseline from always predicting "no relabel", and **AUC is 0.853**: a randomly chosen relabeled SKU gets a higher score than a randomly chosen kept SKU 85% of the time. Pseudo R² and AIC in Part 3 describe fit on the data used to estimate the model. These test numbers describe performance on 450 SKUs the model never saw, which is what matters for next cycle. The test AUC is not lower than the training AUC (0.821), so with seven parameters and 1,050 rows there is no sign of overfitting. The weak spot is recall: at 0.5 the model catches 64% of actual relabels (64 missed). Because a missed correction costs far more than an unnecessary one (Part 5), a lower cutoff would suit this decision better than 0.5.

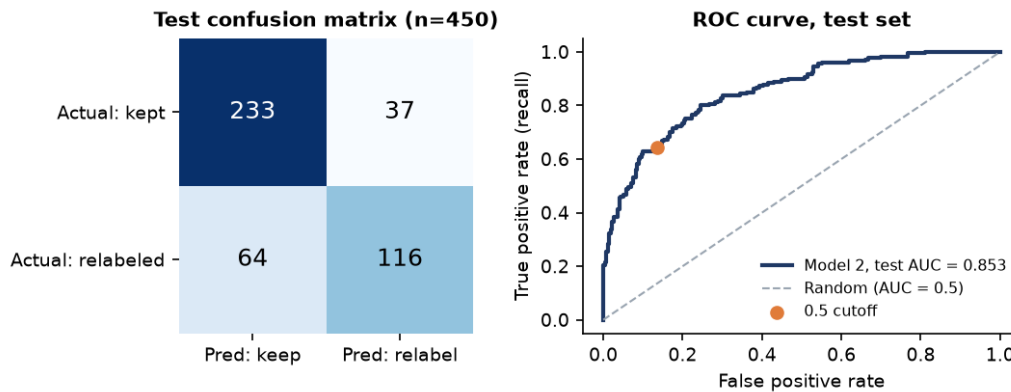


Figure 3. Test-set confusion matrix at a 0.5 cutoff and ROC curve.

Part 5. Economic decision analysis

For each test SKU, $\text{ExpectedWeeklyLoss} = (\text{AbsGapPct}/100) \cdot \text{StorePrice} \cdot \text{WeeklyUnitsSold}$ and $\text{NetBenefit} = \text{ExpectedWeeklyLoss} - \text{RelabelCost}$. **98.7% of test SKUs (444 of 450) recoup the relabel cost within one week** of corrected pricing (98.0% of paper-tag SKUs, 100% of ESL SKUs). The median item loses \$43.18 a week to misalignment against a tag cost of about \$1, so the median payback is 2 trading hours. The 6 exceptions are all cheap, paper-tagged Grocery, Household and PersonalCare items priced within 3.5% of competitors. Yet the store relabeled only 41% of the SKUs that would pay back; the 264 it left alone carry \$22,342 of weekly net benefit in this 30% sample. The benefit is concentrated: the top 20% of SKUs hold 73% of it.

Test segment	SKUs	Median NetBenefit/wk	Total NetBenefit/wk	Share of total	Actually relabeled
Electronics	60	\$284	\$32,034	53%	45%
Apparel	85	\$111	\$13,545	22%	40%
Household	80	\$35	\$6,268	10%	31%
Grocery	150	\$21	\$5,347	9%	46%
PersonalCare	75	\$28	\$3,537	6%	33%
Velocity: Slow third	152	\$24	\$11,518	19%	35%
Velocity: Medium third	150	\$41	\$18,081	30%	43%
Velocity: Fast third	148	\$89	\$31,132	51%	42%

Caveat on the loss formula. ExpectedWeeklyLoss charges the whole gap on every unit sold. That is close to right for an underpriced item (margin given away) but overstates the loss for an overpriced one, where the true cost is lost volume and depends on how price-sensitive customers are. It also ignores the labor needed to relabel hundreds of items in one cycle. The absolute dollar figures are therefore upper bounds; the ranking of SKUs is the more reliable output.

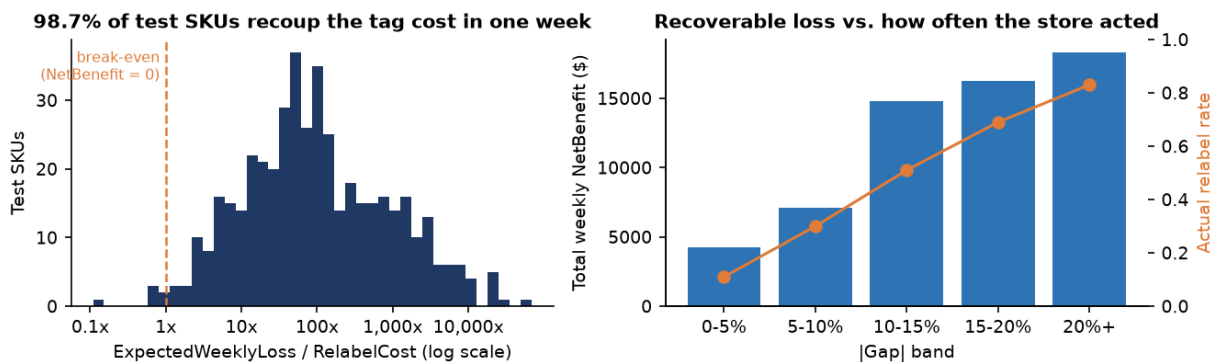


Figure 4. Weekly loss relative to tag cost (left) and total net benefit vs. actual relabel rate by [gap] band (right).

Memo

To: Owner, SunCoast Retail Mart

From: Giuseppe Laricchia, pricing analytics

Date: September 29, 2026

Re: Which price tags to change first, and whether to expand electronic shelf labels

Last cycle we changed 40% of our 1,500 price tags. A model of those decisions, checked on items it had not seen (78% accuracy, AUC 0.85), shows that we reprice when an item drifts away from BigBox Depot, ValueMart and QuickShop, when a supplier raises its cost, when a tag has been unchanged for a long time, and when the item sells quickly. We are held back by the cost of changing a paper tag. Items priced too low are as mispriced as items priced too high and deserve the same attention. The problem is that we act too rarely: about 99% of items would earn back the cost of a new tag within a week, but we changed only 41% of them.

(a) Which items to relabel first

- **Any item more than 10% away from the competitor average, above or below.** These items hold 81% of the recoverable weekly loss.
- **Fast movers within that group first.** Items that are 10%+ misaligned and in our fastest-selling third make up 17% of items but 41% of the benefit, yet only 60% of them were changed last cycle.
- **Electronics, then Apparel.** Higher prices make each point of misalignment expensive: Electronics is 53% of the recoverable loss with 13% of the items; Apparel adds 22%.
- **Items with a recent supplier cost increase or a tag older than about six months,** to protect margin.
- Leave alone cheap, slow items within about 3% of competitors; they are the only ones where the tag costs more than a week of mispricing.

(b) The business case for more electronic labels

An electronic label cuts the cost of a change from about \$1.45 to \$0.06. The direct saving on tags is small, roughly \$382 a cycle across the store. The real value is that cheap changes get made. Re-scoring our paper-tag items with an electronic-label cost, the model expects their relabel rate to rise from 31% to 51%, which is about 206 more corrections per cycle across our 1,045 paper-tag items. Valued at each item's weekly misprice loss, that is up to about \$24,925 a week (about \$23.85 per converted label). Even if the true loss is only a tenth of this upper bound, each label would recover about \$2.39 a week. Convert Electronics and Apparel shelves first, where the value per label is highest, and treat the rollout as a test (see below). We should compare these figures with the vendor's installed price per label before committing.

(c) Caution: these results are associations, not proof of cause

These data record what happened in one pricing cycle; we did not run an experiment. The electronic labels may have been installed first on shelves that already change more often, and fast-moving or older tags may differ in ways the data do not capture. So the effects above describe patterns in our decisions, not guaranteed results of changing them, and the model describes what we did, not what we should have done. The dollar estimates also assume every point of misalignment costs sales or margin on every unit, which overstates the loss on overpriced items. Before a store-wide rollout, convert a random set of paper-tag shelves (for example, half of the Electronics aisles), leave comparable shelves on paper, and compare relabel rates and gross margin over two or three cycles.